

Joshua Cook

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Education

University of British Columbia

Bachelor of Applied Science, Engineering Physics

September 2025 - Present

Projects

Audio to Bluetooth PCB

April 2026 - Present

- Designed an STM32 based 4-layer PCB to stream audio from my digital piano to my Bluetooth headphones
- Completed full PCB design process including: selecting components, creating a schematic, designing a layout, generating Gerber files, preparing, and ordering bill of materials

Reflow toaster oven

June 2026 - July 2026

- Designed and built a programmable reflow oven from a toaster oven to solder surface-mounted components
- Integrated a solid-state relay, Arduino, thermocouple, and an SPI converter to control temperature
- Programmed PID firmware and a front end in Python/Qt that communicate over USB serial

Piano action CAD

Jan 2026 - March 2026

- Modeled a piano action assembly in Fusion 360 to study the piano action design
- Re-create action with correct geometry using publicly available videos, pictures, and repair manuals
- Used Fusion 360 assemblies, joints, and canvases to produce an accurate CAD model

8-bit breadboard computer

October 2024 - June 2025

- Designed and built a simple 8-bit breadboard computer using elementary components
- Designed a simulated an 8-bit computer using a basic logic gate simulator before translating the design into a bill of materials and physical reproduction
- Utilized digital logic design, troubleshooting, and physical troubleshooting through the design process

Work experience

Painter, Magellan Coatings

Jun 2026 - Present

- Work in a team to paint exteriors and interiors

PhotoLab & Tech Specialist, London Drugs

Aug 2024 - Sep 2025

- Received recognition for great customer service and organizational skills
- Operated, maintained, and troubleshooted printing and scanning equipment
- Provided customers with advice and assistance regarding items in the tech department

Skills

Fabrication: CAD (Fusion 360), 3D Printing, Equipment Maintenance, Soldering

Electrical & Systems: PCB Design, Digital Logic, Arduino, C Programming, Troubleshooting, Oscilloscopes, digital multimeter

Core Competencies: Reading Technical Documentation, Fluent in French (Written & Spoken), Level C Standard First Aid